

RACHEL ZOLL

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Hardware prototyping engineer taking **robotic and medical devices** from **concept to in vivo trials**: mechanism design, fabrication, PCBs, firmware, and test. Also experienced in **technical due diligence and commercialization** of deep-tech hardware.

EDUCATION

Harvard University, Cambridge, MA

11/2025

Ph.D., Engineering Sciences — Microrobotics Laboratory · NSF Graduate Research Fellow

Thesis: Millimeter-Scale Multi-Material Manufacturing for Biomedical Devices — Advisor: Prof. Robert J. Wood

University of California, Berkeley, Berkeley, CA

05/2019

M.S., Electrical Engineering & Computer Science — Focus: MEMS, Digital ASIC Design

University of California, Berkeley, Berkeley, CA

05/2018

B.S., Electrical Engineering & Computer Science — Coursework: 28nm SoC for IoT, Analog & Digital Integrated Circuits, Control Systems, Power Electronics, Electronic Medical Devices

TECHNICAL SKILLS

Mechanical Design & Prototyping: Mechanism design; CAD (SolidWorks, Onshape); DFM/DFX; tolerance analysis; soft elastomer fabrication & molding; machining; 3D printing; mechanical testing (Instron).

Micro-Fabrication: MEMS design, simulation & fabrication; laser micromachining; millimeter-scale multi-material manufacturing.

Electronics & Circuit Design: PCB & flex PCB design (Altium, KiCad); digital design (Verilog, CHISEL, Xilinx FPGA); 28nm ASIC tapeout (layout, synthesis, verification; Cadence); LTSpice/SPICE.

Embedded Systems & Controls: C, Python, MATLAB, Assembly, Git; STM32/microcontrollers; firmware debugging (GDB); I2C, SPI, UART, CAN; PID & feedback control; digital signal processing.

Test, Sensing & Validation: Sensor integration (IMU, load cells, force/pressure transducers); hardware-in-the-loop (HIL) validation & custom DAQ software; sensor characterization (accuracy, noise, repeatability); oscilloscopes, parameter & logic analyzers; ex vivo & in vivo testing.

RESEARCH EXPERIENCE

Harvard University — Graduate Student Researcher, Microrobotics Lab

08/2019 – 10/2025

Advisor: Prof. Robert J. Wood

- **Mechanism Design:** Designed, laser-micromachined, and tested a spring-driven sampling mechanism for autonomous GI capsules.
- **Soft-Tissue Anchoring:** Developed a soft-elastomer mechanism for fastener-free anchoring to soft tissue, validated via Instron testing.
- **Flex-Circuit Integration:** Designed a custom flex PCB for system integration of a 32-channel skin-surface sensor array with COTS cable assemblies; validated via impedance testing, electrophysiology testing, and initial ex vivo trials.
- **Test Infrastructure:** Built HIL benchtop characterization suites with custom DAQ software for multi-domain (pressure, flow, thermal-signature) validation of custom micro-machined devices.

UC Berkeley — Graduate & Undergraduate Researcher, Pister, Muller & Maharbiz Labs

08/2017 – 05/2019

- **ASIC Tapeout:** Designed and FPGA-verified a GFSK modulator, CDR, and image-reject filter as part of a team ASIC tapeout of a 28nm FDSOI Bluetooth transceiver ("Tapeout Class").
- **Radio Baseband Design:** Developed and verified a GFSK modulator (FPGA & ASIC) for a crystal-free 2.4GHz single-chip radio (SCuM).
- **Embedded Firmware:** Optimized ARM Thumb assembly and modeled a MEMS finite-state machine for planar silicon motor operation.
- **MEMS Actuators:** Simulated, fabricated, and tested MEMS actuators for parallel insertion of carbon-fiber neural recording electrodes.

PROFESSIONAL EXPERIENCE

Harvard Office of Technology Development & ARPA-Health — Research Analyst, Cambridge, MA

06/2023 – 04/2026

- **Technical Due Diligence:** Evaluated 15+ hardware/deep-tech innovations across a \$50M+ portfolio for technical merit and market potential, resulting in 2 industry licensing leads.
- **Commercialization Strategy:** Partnered with Harvard OTD's Business Development team to build an investor/industry-facing pitch deck supporting a Harvard technology's licensing and go-to-market strategy.

MGH & BWH Hospitals — Robotics Hardware Lead, MESH Innovation Teams

02/2020 – 06/2024

- **Validation Platform:** Designed and deployed a hardware-in-the-loop (HIL) pressure-measurement validation platform used to qualify 3 device iterations; wrote embedded firmware for sensor readout and characterized measurement accuracy ($\pm 1-5\%$) and repeatability to establish a process-control standard for device validation.
- **Prototyping & In Vivo Trials:** Led end-to-end industrialization of an autonomous feeding tube device: defined hardware requirements, managed a 5-person cross-functional team, and executed 2 in vivo swine trials to establish the first gold-standard placement-accuracy benchmark for a procedure with no prior clinical reference data.
- **DFM & Assembly:** Applied DFM/DFX across 5+ design iterations, cutting part count by 50% and kit assembly time from ~20 to ~5 minutes.

Facebook — Operations & Manufacturing Intern, Menlo Park, CA

05/2017 – 08/2017

- Designed and tested hardware interfaces for rapid diagnostic testing of server-rack power units.

Intel Corporation — SSD Failure Analysis Intern, Folsom, CA

05/2016 – 08/2016

- Automated electrical failure-analysis workflows for PCIe SSD products and trained colleagues in wafer processing.

FMC Schilling Robotics — Electrical Engineering Intern, Davis, CA

05/2015 – 08/2015

- Tuned PID control of a high-pressure subsea ROV hydraulic valve pack to minimize overshoot and prevent valve overload.

PUBLICATIONS

12+ peer-reviewed papers in PNAS Nexus, IROS, IEEE Transactions on NanoBioscience, VLSI Symposium, and ISCAS. Full list on request.

HONORS & FELLOWSHIPS

- NSF Graduate Research Fellowship, \$53,000/yr (2019 – 2022)
- HealthTech Fellowship Finalist, Harvard Medical School (2023)
- Certificate of Distinction in Teaching, Harvard Bok Center for Teaching and Learning (2020)
- EECS Departmental Fellowship & EECS Excellence Award, UC Berkeley (2018)
- Regents' and Chancellor's Scholarship, UC Berkeley, \$25,000/yr tuition fellowship (2014 – 2018)

TEACHING & LEADERSHIP

- Teaching Fellow, GENED 1080: Engineering the Acoustical World, Harvard University (Fall 2020, Fall 2023)
- Discussion TA, EE16A: Designing Information Devices and Systems I, UC Berkeley (Spring & Fall 2017)
- Lab Assistant, EE16A: Designing Information Devices and Systems I, UC Berkeley (Fall 2015)
- Harvard SEAS Graduate Council, Co-President (2020 – 2022); MIT Sidney Pacific Graduate Dorm Executive Council & Trustee (2021 – 2023)

LANGUAGES

Mandarin Chinese (Intermediate); French (Intermediate).